We create chemistry

Safety data sheet

1. Substance/preparation and manufacturer/supplier identification

Product name:
Frivon

Use: crop protection product, fungicide

Manufacturer/supplier:
BASF Philippines, Inc.
Upper Penthouse CTP ASEAN Tower
Asean Drive, Spectrum District
Filinvest Corporate City, Alabang,
Muntinlupa City, 1781, Metro Manila
PHILIPPINES
Telephone: +63 2 8811-8001
E-mail address: psr.ph@basf.com

Emergency information:
National emergency number:
+63 2 8831 5576
International emergency number:
Telephone: +49 180 2273-112

2. Hazard identification

Classification of the substance and mixture:
Reproductive toxicity: Cat.2 (unborn child)
Hazardous to the aquatic environment - acute: Cat.1
Hazardous to the aquatic environment - chronic: Cat.1

Label elements and precautionary statement:

Pictogram:

Signal Word:
Warning

Hazard Statement:

H361	Suspected of damaging the unborn child.
H400	Very toxic to aquatic life.
H410	Very toxic to aquatic life with long lasting effects.

Precautionary Statement:

P101	If medical advice is needed, have product container or label at hand.
P102	Keep out of reach of children.
P103	Read carefully and follow all instructions.

Precautionary Statements (Prevention):

P280	Wear protective gloves, protective clothing and eye protection or face protection.
P201	Obtain special instructions before use.
P202	Do not handle until all safety precautions have been read and understood.

Precautionary Statements (Response):

P391	Collect spillage.
P308 + P313	IF exposed or concerned: Get medical attention.

Precautionary Statements (Storage):

P405	Store locked up.

Precautionary Statements (Disposal):

P501	Dispose of contents and container to hazardous or special waste collection point.

Other hazards which do not result in classification:
See section 12 - Results of PBT and vPvB assessment.
If applicable information is provided in this section on other hazards which do not result in classification but which may contribute to the overall hazards of the substance or mixture.

May produce an allergic reaction. Contains:
HEXAMETHYLENE DIISOCYANATE, OLIGOMERS, 2,2'-iminodi(ethylamine), 1,2-benzisothiazol-3(2H)-one, 2-methyl-2H-isothiazol-3-one

3. Composition/information on ingredients

Chemical nature

Substance nature: mixture

crop protection product, fungicide, capsule suspension (CS)

Hazardous ingredients

Carbamic acid, [2-[[[1-(4-chlorophenyl)-1H-pyrazol-3-yl]oxy]methyl]phenyl]methoxy-, methyl ester

Content (W/W): 9.5 %	Acute Tox.: Cat. 3 (Inhalation - mist)
CAS Number: 175013-18-0	Acute Tox.: Cat. 4 (oral)
	Skin Irrit.: Cat. 2
	Repr.: Cat. 2 (unborn child)
	STOT SE: Cat. 3 (irr. to respiratory syst.)
	STOT RE (Liver, Nasal cavity, Gastrointestinal tract): Cat. 2
	Aquatic Acute: Cat. 1
	Aquatic Chronic: Cat. 1
	M-factor acute: 100
	M-factor chronic: 100

Alcohols, C8-C10, ethoxylated, propoxylated (polymer)

Content (W/W): < 15 %	Acute Tox.: Cat. 5 (oral)
CAS Number: 68603-25-8	Eye Dam./Irrit.: Cat. 2A
	Skin Corr./Irrit.: Cat. 2
	Aquatic Acute: Cat. 3

solvent naphtha

Content (W/W): < 15 %	Asp. Tox.: Cat. 1
CAS Number: 64742-94-5	Aquatic Acute: Cat. 2
	Aquatic Chronic: Cat. 2

Glycerol

Content (W/W): < 10 %	Acute Tox.: Cat. 5 (Inhalation - vapour)
CAS Number: 56-81-5	

(OLIGOMER) Hexamethylene diisocyanate isocyanurate-type oligomers

Content (W/W): < 5 %	Acute Tox.: Cat. 4 (Inhalation - mist)
CAS Number: 28182-81-2	Skin Sens.: Cat. 1
	STOT SE: Cat. 3 (irr. to respiratory syst.)

Alcohols, C12-18, ethoxylated propoxylated

Content (W/W): < 5 %	Aquatic Acute: Cat. 2
CAS Number: 69227-21-0	

2,2'-iminodi(ethylamine)

Content (W/W): < 1 %	Acute Tox.: Cat. 4 (oral)
CAS Number: 111-40-0	Acute Tox.: Cat. 2 (Inhalation - mist)
	Acute Tox.: Cat. 4 (dermal)
	Skin Corr.: Cat. 1B
	Eye Dam.: Cat. 1
	Skin Sens.: Cat. 1B
	STOT SE: Cat. 3 (irr. to respiratory syst.)
	Aquatic Acute: Cat. 3

1,2-benzisothiazol-3(2H)-one

Content (W/W): < 0.01 %	Acute Tox.: Cat. 2 (Inhalation - dust)
CAS Number: 2634-33-5	Acute Tox.: Cat. 4 (oral)
	Skin Irrit.: Cat. 2
	Eye Dam.: Cat. 1
	Skin Sens.: Cat. 1A
	Aquatic Acute: Cat. 1
	Aquatic Chronic: Cat. 1
	M-factor acute: 1
	M-factor chronic: 1

2-methyl-2H-isothiazol-3-one

Content (W/W): < 0.01 %	Acute Tox.: Cat. 2 (Inhalation - dust)
CAS Number: 2682-20-4	Acute Tox.: Cat. 3 (oral)
	Acute Tox.: Cat. 3 (dermal)
	Skin Corr.: Cat. 1B
	Eye Dam.: Cat. 1
	Skin Sens.: Cat. 1A
	Aquatic Acute: Cat. 1
	Aquatic Chronic: Cat. 1
	M-factor acute: 10
	M-factor chronic: 1

4. First-Aid Measures

General advice:
Remove contaminated clothing.

If inhaled:
Keep patient calm, remove to fresh air, seek medical attention.

On skin contact:
Wash thoroughly with soap and water

On contact with eyes:
Wash affected eyes for at least 15 minutes under running water with eyelids held open.

On ingestion:
Immediately rinse mouth and then drink 200-300 ml of water, seek medical attention.

Note to physician:
Symptoms: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11., (Further) symptoms and / or effects are not known so far
Hazards: Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11. (Further) symptoms and / or effects are not known so far
Treatment: Treat according to symptoms (decontamination, vital functions), no known specific antidote.

5. Fire-Fighting Measures

Suitable extinguishing media:
dry powder, foam, water spray, carbon dioxide

Specific hazards:
carbon monoxide, carbon dioxide, hydrogen chloride, nitrogen oxides, halogenated compounds, sulfur oxides, silica compounds, isocyanate
The substances/groups of substances mentioned can be released in case of fire.

Special protective equipment:
Wear self-contained breathing apparatus and chemical-protective clothing.

Further information:
Collect contaminated extinguishing water separately, do not allow to reach sewage or effluent systems. Dispose of fire debris and contaminated extinguishing water in accordance with official regulations. In case of fire and/or explosion do not breathe fumes. Keep containers cool by spraying with water if exposed to fire.

6. Accidental Release Measures

Personal precautions:
Do not breathe vapour/spray.Use personal protective clothing.Avoid contact with the skin, eyes and clothing.

Environmental precautions:
Do not discharge into the subsoil/soil.Do not discharge into drains/surface waters/groundwater.

Methods for cleaning up or taking up:
For small amounts: Pick up with suitable absorbent material (e.g. sand, sawdust, general-purpose binder, kieselguhr).
For large amounts: Dike spillage. Pump off product.
Dispose of absorbed material in accordance with regulations. Collect waste in suitable containers, which can be labeled and sealed. Clean contaminated floors and objects thoroughly with water and detergents, observing environmental regulations. Wear suitable protective equipment.

7. Handling and Storage

Handling

No special measures necessary if stored and handled correctly. Ensure thorough ventilation of stores and work areas. When using do not eat, drink or smoke. Hands and/or face should be washed before breaks and at the end of the shift.

Protection against fire and explosion:
No special precautions necessary. The substance/product is non-combustible. Product is not explosive.

Storage

Segregate from foods and animal feeds.
Further information on storage conditions: Keep away from heat. Protect from direct sunlight.

Storage stability:
Storage duration: 36 Months

Protect from temperatures below: 0 °C
Changes in the properties of the product may occur if substance/product is stored below indicated temperature for extended periods of time.
Protect from temperatures above: 35 °C
Changes in the properties of the product may occur if substance/product is stored above indicated temperature for extended periods of time.

8. Exposure controls and personal protection

Components with occupational exposure limits

solvent naphtha, 64742-94-5;
Skin Designation (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Danger of cutaneous absorption
TWA value 200 mg/m3 (ACGIHTLV), Non-aerosol
Measured as: total hydrocarbon vapor
Application restricted to conditions in which there are negligible aerosol exposures.

Pyraclostrobin, 175013-18-0;
TWA value 0.13 mg/m3 (BASF recomm. occupational exposure limit)

Naphthalene, 1-methyl-, 90-12-0;
CLV 0.2 mg/m3 (OEL (PH)), Particulate
Measured as: benzene solubles
Skin Designation (ACGIHTLV)
Danger of cutaneous absorption
TLV-SL (ACGIHTLV)
TWA value 0.05 ppm (ACGIHTLV)

Naphthalene, 2-methyl-, 91-57-6;
TWA value 0.5 ppm (ACGIHTLV)
Skin Designation (ACGIHTLV)
The substance can be absorbed through the skin.
CLV 0.2 mg/m3 (OEL (PH)), Particulate
Measured as: benzene solubles
Skin Designation (ACGIHTLV)
Danger of cutaneous absorption
TLV-SL (ACGIHTLV)
TWA value 0.05 ppm (ACGIHTLV)

Personal protective equipment

Respiratory protection:
Suitable respiratory protection for higher concentrations or long-term effect: Combination filter for gases/vapours of organic, inorganic, acid inorganic and alkaline compounds (e.g. EN 14387 Type ABEK).

Hand protection:
Suitable chemical resistant safety gloves (EN ISO 374-1) also with prolonged, direct contact (Recommended: Protective index 6, corresponding > 480 minutes of permeation time according to EN ISO 374-1): E.g. nitrile rubber (0.4 mm), chloroprene rubber (0.5 mm), butyl rubber (0.7 mm) etc.

Eye protection:
Safety glasses with side-shields (frame goggles) (e.g. EN 166)

Body protection:
Body protection must be chosen depending on activity and possible exposure, e.g. apron, protecting boots, chemical-protection suit (according to EN 14605 in case of splashes or EN ISO 13982 in case of dust).

General safety and hygiene measures:
The statements on personal protective equipment in the instructions for use apply when handling crop-protection agents in final-consumer packing. Wearing of closed work clothing is recommended. Store work clothing separately. Keep away from food, drink and animal feeding stuffs.

9. Physical and Chemical Properties

Form:	liquid
Colour:	pale beige
Odour:	faintly aromatic
Odour threshold:	Not determined due to potential health hazard by inhalation.
pH value:	approx. 6 - 8 (20 °C)
Melting temperature:	approx. 0 °C Information applies to the solvent.
boiling temperature:	approx. 100 °C Information applies to the solvent.
Flash point:	Non-flammable.
Evaporation rate:	not applicable
Flammability (solid/gas):	not applicable
Lower explosion limit:	As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.

Upper explosion limit:	As a result of our experience with this product and our knowledge of its composition we do not expect any hazard as long as the product is used appropriately and in accordance with the intended use.	
Ignition temperature:	approx. 438 °C	(Directive 92/69/EEC, A.15)
Thermal decomposition:	145 °C , 130 kJ/kg 360 °C , 160 kJ/kg Not a substance liable to self-decomposition according to UN transport regulations, class 4.1.	
Explosion hazard:	Based on the chemical structure there is no indication of explosive properties.	(Directive 92/69/EEC, A.14)
Fire promoting properties:	not fire-propagating	(Directive 2004/73/EC, A.21)
Vapour pressure:	approx. 23 hPa (20 °C) Information applies to the solvent.	
Density:	approx. 1.05 g/cm3 (20 °C)	
Relative vapour density (air):	not applicable	
Solubility in water:	dispersible	
Partitioning coefficient n-octanol/water (log Pow):	not applicable for mixtures	
Viscosity, dynamic:	approx. 271 mPa.s (20 °C, 100 1/s)	

Other Information:
If necessary, information on other physical and chemical parameters is indicated in this section.

Particle characteristics

Particle size distribution: The substance / product is marketed or used in a non solid or granular form. -

10. Stability and Reactivity

Conditions to avoid:
See SDS section 7 - Handling and storage.

Thermal decomposition:	145 °C, 130 kJ/kg
Thermal decomposition:	360 °C, 160 kJ/kg

Thermal decomposition:	Not a substance liable to self-decomposition according to UN transport regulations, class 4.1.

Substances to avoid:
strong oxidizing agents, strong bases, strong acids

Hazardous reactions:
No hazardous reactions if stored and handled as prescribed/indicated.

Hazardous decomposition products:
No hazardous decomposition products if stored and handled as prescribed/indicated.

Chemical stability:
The product is stable if stored and handled as prescribed/indicated.

Reactivity:
No hazardous reactions if stored and handled as prescribed/indicated.

11. Toxicological Information

Routes of exposure

Acute oral toxicity

Experimental/calculated data:
LD50rat (oral): > 2,000 mg/kg (OECD Guideline 401)
No mortality was observed.

Acute inhalation toxicity

LC50 rat (by inhalation): > 2.4 mg/l (OECD Guideline 403)
Highest concentration technically achievable. No mortality was observed.

Acute dermal toxicity

LD50 rat (dermal): > 5,000 mg/kg (OECD Guideline 402)

Assessment of acute toxicity

Virtually nontoxic after a single ingestion. Virtually nontoxic by inhalation. Virtually nontoxic after a single skin contact.

Symptoms

Information, i.e. additional information on symptoms and effects may be included in the GHS labeling phrases available in Section 2 and in the Toxicological assessments available in Section 11.
(Further) symptoms and / or effects are not known so far

Irritation

Assessment of irritating effects:
Not irritating to eyes and skin.

Experimental/calculated data:
Skin corrosion/irritation rabbit: non-irritant

Serious eye damage/irritation rabbit: non-irritant

Respiratory/Skin sensitization

Assessment of sensitization:
No sensitizing effect.

Experimental/calculated data:
modified Buehler test guinea pig: Non-sensitizing.

Germ cell mutagenicity

Assessment of mutagenicity:
The product has not been tested. The statement has been derived from the properties of the individual components. Mutagenicity tests revealed no genotoxic potential.

Carcinogenicity

Assessment of carcinogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: 2,2'-iminodi(ethylamine)
Assessment of carcinogenicity:
The substance showed no carcinogenic activity in animals after chronic administration to the skin.

Under certain conditions the substance can form nitrosamines. Nitrosamines are carcinogenic in animal studies.

Reproductive toxicity

Assessment of reproduction toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components. The results of animal studies gave no indication of a fertility impairing effect.

Developmental toxicity

Assessment of teratogenicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Assessment of teratogenicity:
Indications of possible developmental toxicity/teratogenicity were seen in animal studies.

Specific target organ toxicity (single exposure)

Based on the available information there is no specific target organ toxicity to be expected after a single exposure.

Remarks: The product has not been tested. The statement has been derived from the properties of the individual components.

Repeated dose toxicity and Specific target organ toxicity (repeated exposure)

Assessment of repeated dose toxicity:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Assessment of repeated dose toxicity:
Repeated exposure may affect certain organs. Target organs: Liver, gastrointestinal tract and nasal cavity

Information on: 2,2'-iminodi(ethylamine)
Assessment of repeated dose toxicity:
The substance may cause damage to the liver after repeated ingestion of high doses, as shown in animal studies. The substance may cause damage to the kidney after repeated ingestion of high doses, as shown in animal studies. The product has not been fully tested. The statements have been derived in parts from products of a similar structure or composition.

Information on: (OLIGOMER) Hexamethylene diisocyanate isocyanurate-type oligomers
Assessment of repeated dose toxicity:
After repeated exposure the prominent effect is local irritation.

Aspiration hazard

not applicable

Other relevant toxicity information

Misuse can be harmful to health.

12. Ecological Information

Ecotoxicity

Assessment of aquatic toxicity:
Very toxic to aquatic life with long lasting effects.

Toxicity to fish:
LC50 (96 h) > 1.06 mg/l, Cyprinus carpio (OECD 203; ISO 7346; 92/69/EWG, C.1, static)

Aquatic invertebrates:
EC50 (48 h) 0.624 mg/l, Daphnia magna

Aquatic plants:
EC10 (72 h) 7.7 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201)

EC50 (72 h) 27.7 mg/l (growth rate), Pseudokirchneriella subcapitata (OECD Guideline 201)

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Chronic toxicity to fish:

No observed effect concentration (98 d) approx. 0.00235 mg/l, Oncorhynchus mykiss (OECD Guideline 210, Flow through.)

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Chronic toxicity to aquatic invertebrates:
No observed effect concentration (21 d), 0.004 mg/l, Daphnia magna (OECD Guideline 202, part 2, semistatic)
The details of the toxic effect relate to the nominal concentration.

No observed effect concentration (31 d), 0.000365 mg/l, Mysidopsis bahia
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Mobility

Assessment transport between environmental compartments:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Assessment transport between environmental compartments:
Following exposure to soil, adsorption to solid soil particles is probable, therefore contamination of groundwater is not expected.

Persistence and degradability

Assessment biodegradation and elimination (H2O):
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
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Bioaccumulation potential

Assessment bioaccumulation potential:
The product has not been tested. The statement has been derived from the properties of the individual components.

Information on: pyraclostrobin (ISO); methyl N-{2-[1-(4-chlorophenyl)-1H-pyrazol-3-yloxymethyl]phenyl}(N-methoxy)carbamate
Bioaccumulation potential:
Bioconcentration factor: 379 - 507, Oncorhynchus mykiss (OECD Guideline 305)
Accumulation in organisms is not to be expected.

Additional information

Other ecotoxicological advice:
Do not discharge product into the environment without control.

13. Disposal Considerations

Must be sent to a suitable incineration plant, observing local regulations.

Contaminated packaging:
Contaminated packaging should be emptied as far as possible and disposed of in the same manner as the substance/product.

14. Transport Information

Domestic transport:

UN number or ID number:	UN 3082
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (PYRACLOSTROBIN)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Sea transport
IMDG

UN number or ID number:	UN 3082
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (PYRACLOSTROBIN)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes Marine pollutant: YES
Special precautions for user:	EmS: F-A; S-F

Air transport
IATA/ICAO

UN number or ID number:	UN 3082
UN proper shipping name:	ENVIRONMENTALLY HAZARDOUS SUBSTANCE, LIQUID, N.O.S. (PYRACLOSTROBIN)
Transport hazard class(es):	9, EHSM
Packing group:	III
Environmental hazards:	yes
Special precautions for user:	None known

Maritime transport in bulk according to IMO instruments

Maritime transport in bulk is not intended.

Further information
Product may be shipped as non-hazardous in suitable packages containing a net quantity of 5 L or less under the provisions of various regulatory agencies: ADR, RID, ADN: Special Provision 375; IMDG: 2.10.2.7; IATA: A197; TDG: Special Provision 99(2); 49CFR: §171.4 (c) (2) and also the Special Provision 375 in Appendix B which is regulated in China "Regulations Concerning Road Transportation of Dangerous Goods Part 3: Index of dangerous goods name and transportation requirements" (JT/T 617.3)

15. Regulatory Information

Other regulations

To avoid risks to man and the environment, comply with the instructions for use.

1. Joint DTI-DENR-DA-DOF-DOH-DILG-DOLE-DOTC Administrative Order No. 01 Series of 2009 on “The Adoption and Implementation of the Globally Harmonized System of Classification and Labeling of Chemicals (GHS)”

2. DAO 2015-09 "Rules and Procedures for the Implementation of the Globally Harmonized System (GHS) of Classification and Labelling of Chemicals in Prepration of Safety Data Sheet (SDS) and Labelling Requirements of Toxic Chemical Substances”

3. Republic Act No. 6969, “Toxic Substances and Hazardous and Nuclear Wastes Control Act of 1990”

The regulatory information is not intended to be comprehensive. Other regulations may apply to the material

Registration status:

PICCS, PH

Contains non-registered, non-listed substance., Individual registration may be required.

16. Other Information

Vertical lines in the left hand margin indicate an amendment from the previous version.

The data contained in this safety data sheet are based on our current knowledge and experience and describe the product only with regard to safety requirements. This safety data sheet is neither a Certificate of Analysis (CoA) nor technical data sheet and shall not be mistaken for a specification agreement. Identified uses in this safety data sheet do neither represent an agreement on the corresponding contractual quality of the substance/mixture nor a contractually designated use. It is the responsibility of the recipient of the product to ensure any proprietary rights and existing laws and legislation are observed.